VISHAL MART

Full-Stack E-Commerce Web Application

Project Technical Report

Project Name: Vishal Mart
Type: Full-Stack Web Application
Frontend: React.js
Backend: Node.js + Express.js
Database: MongoDB Atlas
Deployment: Vercel (Frontend + Backend)
Version Control: GitHub

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Vishal Mart is a modern, full-stack e-commerce web application designed to provide a seamless online shopping experience. The platform supports customer registration, product browsing, cart management, order placement, and an admin dashboard for complete store management.

The application was built using the MERN-like stack (React + Node.js + MongoDB) and deployed on Vercel with MongoDB Atlas as the cloud database. The project demonstrates real-world deployment practices including serverless architecture, JWT authentication, and RESTful API design.

- React.js 19 — UI framework with hooks and context API
- React Router DOM v7 — Client-side routing and navigation
- Axios — HTTP client for API communication
- HTML5 QR Code — QR code scanning functionality
- Socket.io Client — Real-time communication
- CRACO — Create React App Configuration Override

Backend

- Node.js 24.x — JavaScript runtime environment
- Express.js 5 — Web application framework
- Mongoose — MongoDB ODM for schema modeling
- JWT (jsonwebtoken) — Stateless authentication tokens
- bcryptjs — Password hashing and verification
- CORS — Cross-Origin Resource Sharing middleware
- dotenv — Environment variable management

Database & Infrastructure

- MongoDB Atlas — Cloud-hosted NoSQL database (M0 free tier)
- Vercel — Serverless deployment platform (frontend + backend)
- GitHub — Version control and source code repository

The application follows a decoupled client-server architecture deployed as two separate Vercel projects:

- Frontend (React SPA) !’ deployed at frontend-black-sigma-94.vercel.app
- Backend (Express API) !’ deployed at backend-eight-chi-55.vercel.app
- Database !’ MongoDB Atlas cluster (cluster0.v4ykp90.mongodb.net)
- Frontend communicates with backend via HTTPS REST API calls using Axios
- Backend connects to MongoDB Atlas using Mongoose with serverless connection caching
- JWT tokens stored in localStorage for stateless session management
- All API routes protected with authMiddleware verifying JWT on each request

Request Flow:
User Browser !’ React Frontend !’ Axios HTTP !’ Vercel Serverless Function !’ Express Router !’ Controller !’ Mongoose !’ MongoDB Atlas

- User Registration with email, phone, address, and password
- Secure Login with JWT token-based authentication
- Browse products by category with search and filter
- Add to Cart with quantity management
- Place orders and view order history with status tracking
- QR Code scanning for quick product lookup
- Profile management — update name, phone, address
- Change password with current password verification
- OTP-based password recovery (send/verify OTP)
- Feedback submission system

Admin Features

- Secure admin login with role-based access control
- Admin Dashboard with sales overview and statistics
- Product management — add, view, and delete products
- Order management — view all orders, update order status
- User management — view all registered customers
- Seed database with initial products via /api/seed endpoint

MongoDB Atlas is used as the database with Mongoose ODM. Three main collections are defined:

Users Collection

Field	Type	Description
_id	ObjectId	Auto-generated primary key
name	String (required)	Full name of the user
email	String (unique)	Email address for login
password	String (hashed)	bcrypt hashed password
phone	String	Contact phone number
address	String	Delivery address
role	String (enum)	"user" or "admin" (default: user)
otp	String	OTP for password recovery
otpExpiry	Date	OTP expiration timestamp
createdAt	Date	Auto-managed by Mongoose

Products Collection

Field	Type	Description
_id	ObjectId	Auto-generated primary key
name	String (required)	Product name
category	String (required)	Product category
price	Number (required)	Product price in INR
stock	Number	Available stock quantity
image	String	Product image URL
description	String	Product description text

Orders Collection

Field	Type	Description
_id	ObjectId	Auto-generated primary key
userId	ObjectId (ref: User)	Reference to the ordering user
items	Array	Array of {productId, name, price, qty}
total	Number	Total order amount in INR
status	String (enum)	pending / confirmed / delivered
createdAt	Date	Order placement timestamp

Base URL: https://backend-eight-chi-55.vercel.app/api

Authentication Routes (/api/auth)

Method + Endpoint	Auth Required	Description
POST /register	No	Register a new user account
POST /login	No	Login and receive JWT token
GET /profile	Yes (JWT)	Get current user profile
PUT /profile	Yes (JWT)	Update user profile details
PUT /change-password	Yes (JWT)	Change user password
GET /users	Yes (Admin)	Get all registered users
POST /send-otp	No	Send OTP to email for recovery
POST /verify-otp	No	Verify OTP and reset password

Product Routes (/api/products)

Method + Endpoint	Auth Required	Description
GET /	No	Get all products
POST /	Yes (Admin)	Add a new product
DELETE /:id	Yes (Admin)	Delete a product by ID

Order Routes (/api/orders)

Method + Endpoint	Auth Required	Description
POST /	Yes (JWT)	Place a new order
GET /my	Yes (JWT)	Get orders for current user
GET /all	Yes (Admin)	Get all orders (admin view)
PUT /:id/status	Yes (Admin)	Update order status

Utility Routes

Method + Endpoint	Auth Required	Description
GET /	No	API health check message
GET /api/health	No	Returns { status: "ok" }
GET /api/seed	No	Seeds admin user and products

• Frontend and backend deployed as two separate Vercel projects under vishnu640s-projects organization
• Backend uses @vercel/node runtime — server.js exported as a serverless function
• Frontend uses react-scripts build with cross-env for environment compatibility
• Both projects auto-deploy from GitHub repository on every push to main branch
• Environment variables configured directly in Vercel project settings dashboard

Serverless Architecture

• server.js uses isConnected flag to cache MongoDB connection across warm invocations
• connectDB() called via Express middleware before each request (not at startup)
• No app.listen() in production — Vercel handles the HTTP server lifecycle
• module.exports = app allows Vercel to import and invoke the Express app

MongoDB Atlas Configuration

• Cluster: cluster0.v4ykp90.mongodb.net (M0 Free Tier)
• Database name: vishalmart
• Network Access: 0.0.0.0/0 (allow all IPs — required for Vercel dynamic IPs)
• Connection string uses SRV format: mongodb+srv://...
• Mongoose connection options: retryWrites=true, w=majority

GitHub Repository

• Repository: https://github.com/Vishnu640/vishal-mart
• Monorepo structure with /frontend and /backend directories
• Separate vercel.json in each subdirectory for independent deployments
• .gitignore excludes node_modules, .env files, and build artifacts

Environment Variables (Backend — Vercel)

Variable	Value	Purpose
MONGO_URI	mongodb+srv://...	MongoDB Atlas connection string
JWT_SECRET	Set in Vercel	Secret key for JWT signing
PORT	5001 (local only)	Local development server port

• Password Hashing: All passwords hashed using bcryptjs with salt rounds before storage
• JWT Authentication: Stateless tokens signed with secret key, verified on protected routes
• Role-Based Access Control: Admin routes protected by role check in authMiddleware
• CORS Configuration: Configured to allow cross-origin requests from frontend domain
• Environment Variables: Sensitive data (DB URI, JWT secret) stored in .env, never committed
• Input Validation: Required fields enforced at Mongoose schema level
• OTP Expiry: Password recovery OTPs have expiration timestamps to prevent replay attacks
• No SQL Injection: MongoDB with Mongoose eliminates SQL injection attack surface

```
vishal-mart/
% % %  frontend/
%    % % %  public/             # Static assets
%    % % %  src/
%    %    % % %  api/
%    %    %    % % %  axios.js      # Axios instance with base URL
%    %    % % %  components/
%    %    %    % % %  Navbar.js     # Navigation bar component
%    %    %    % % %  ProductCard.js
%    %    %    % % %  QRScanner.js
%    %    % % %  context/
%    %    %    % % %  AuthContext.js  # Global auth state
%    %    % % %  pages/
%    %        % % %  Home.js       # Product listing page
%    %        % % %  Login.js      # Login form
%    %        % % %  Register.js   # Registration form
%    %        % % %  Cart.js       # Shopping cart
%    %        % % %  Orders.js     # Order history
%    %        % % %  AdminDashboard.js
%    %        % % %  Settings.js   # Profile & password
%    %        % % %  Feedback.js
%    %        % % %  Welcome.js
%    % % %  .env.production      # REACT_APP_API_URL
%    % % %  vercel.json
%
% % %  backend/
    % % %  config/
    %    % % %  db.js            # Mongoose connection
    % % %  controllers/
    %    % % %  authController.js
    %    % % %  productController.js
    %    % % %  orderController.js
    % % %  middleware/
    %    % % %  authMiddleware.js # JWT verification
    % % %  models/
    %    % % %  User.js          # Mongoose User schema
    %    % % %  Product.js       # Mongoose Product schema
    %    % % %  Order.js         # Mongoose Order schema
    % % %  routes/
    %    % % %  authRoutes.js
    %    % % %  productRoutes.js
    %    % % %  orderRoutes.js
    % % %  seeder.js           # Product seed data
    % % %  adminSeeder.js      # Admin user seeder
    % % %  server.js           # Main Express app
    % % %  .env                # Local environment vars
    % % %  vercel.json
```

Service	URL	Status
Frontend	https://frontend-black-sigma-94.vercel.app	Live
Backend API	https://backend-eight-chi-55.vercel.app	Live
GitHub Repo	https://github.com/Vishnu640/vishal-mart	Public
MongoDB Atlas	cluster0.v4ykp90.mongodb.net	Connected

Admin Credentials

Field	Value	Notes
Email	admin@vishalmart.com	Admin login email
Password	Admin@123	Admin login password
Role	admin	Full access to dashboard

11. Challenges & Solutions

Vercel free tier blocks outbound TCP port 3306. Tried Railway, filess.io, Clever Cloud — all blocked. Migrated entire app from Sequelize/MySQL to Mongoose/MongoDB Atlas (port 27017 via SRV DNS — allowed).

%¶ Local Network Blocks MongoDB Port

Local network also blocks port 27017 and SRV DNS resolution. Created /api/seed HTTP endpoint on Vercel to seed data remotely without needing local DB access.

%¶ Serverless Cold Start & Connection Pooling

Mongoose creates new connections on every cold start. Implemented isConnected flag in server.js to cache and reuse existing connections across warm invocations.

%¶ Monorepo Build Issues on Vercel

Single Vercel project could not build both frontend and backend. Deployed as two separate Vercel projects with individual vercel.json configurations.

%¶ Duplicate Variable Declaration Bug

The /api/seed endpoint had duplicate const Product declaration causing a syntax error. Products returned empty array. Fixed by removing the duplicate require statement.

%¶ Cross-Platform Build Scripts

Windows development environment caused issues with environment variable syntax in npm scripts. Added cross-env package to normalize env var syntax across platforms.

Vishal Mart successfully demonstrates a complete, production-ready e-commerce application built with modern web technologies. The project covers the full software development lifecycle — from local development to cloud deployment.

Key achievements of this project:

- Full-stack application with React frontend and Node.js/Express backend
- Cloud database integration with MongoDB Atlas
- Serverless deployment on Vercel with proper connection caching
- JWT-based authentication with role-based access control
- Complete admin dashboard for product, order, and user management
- Real-world problem solving — database migration, network restrictions, deployment issues
- Version control with GitHub and CI/CD via Vercel auto-deploy

The project demonstrates proficiency in:

- React.js — component architecture, hooks, context API, routing
- Node.js & Express.js — RESTful API design, middleware, routing
- MongoDB & Mongoose — schema design, CRUD operations, population
- Authentication — JWT tokens, bcrypt hashing, OTP flows
- Cloud deployment — Vercel, MongoDB Atlas, environment configuration
- DevOps basics — Git, GitHub, environment variables, serverless functions

https://frontend-black-sigma-94.vercel.app